

Analysis Report

Global dataset report

This report is the output of the Amazon SageMaker Clarify analysis. The report is split into following parts:

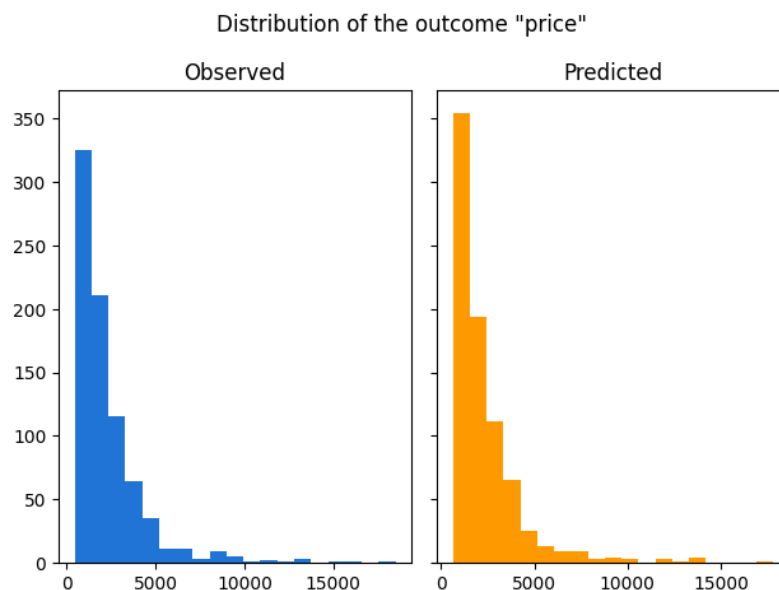
1. Analysis configuration
2. High level model performance
3. Pretraining bias metrics
4. Posttraining bias metrics

Analysis Configuration

Bias analysis requires you to configure the outcome label column, the facet and optionally a group variable. Generating explanations requires you to configure the outcome label. You configured the analysis with the following variables. The complete analysis configuration is appended at the end.

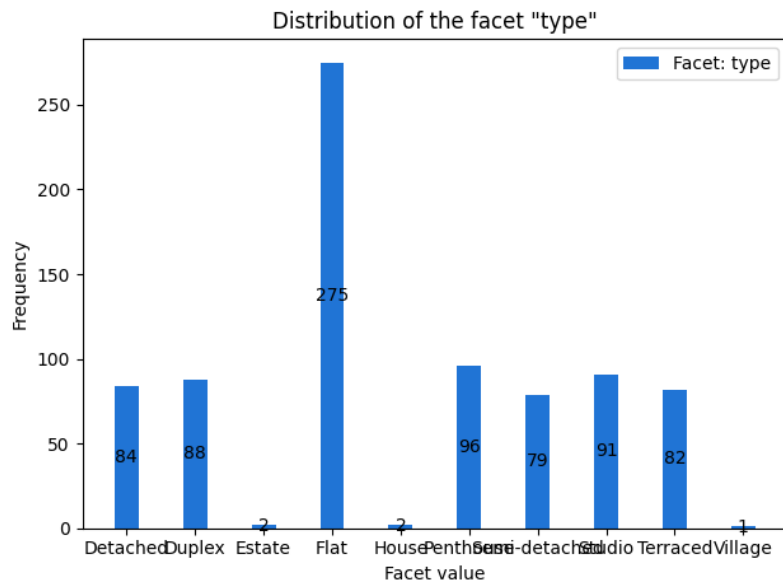
Outcome label: You chose the column `price` in the input data as the outcome label. Bias metric computation requires designating the positive outcome. You chose `price > 1500` as the positive outcome. The outcome label `price` varied between `450.00` and `19500.00`.

The figure below shows the distribution of values of `price`.



Facet: You chose the column `type` in the input data as the facet. `type` consisted of values `['Detached', 'Duplex', 'Estate', 'Flat', 'House', 'Penthouse', 'Semi-detached', 'Studio', 'Terraced', 'Village']`. Bias metrics were computed by comparing the inputs `type = Detached` with all other inputs, then by comparing inputs `type = Duplex` with all other inputs, then by comparing inputs `type = Estate` with all other inputs, then by comparing inputs `type = Flat` with all other inputs, then by comparing inputs `type = House` with all other inputs, then by comparing inputs `type = Penthouse` with all other inputs, then by comparing inputs `type = Semi-detached` with all other inputs, then by comparing inputs `type = Studio` with all other inputs, then by comparing inputs `type = Terraced` with all other inputs, then by comparing inputs `type = Village` with all other inputs.

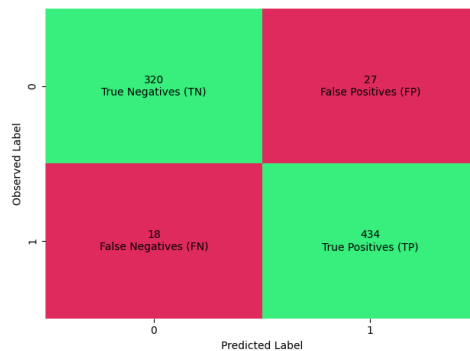
The figure below shows the distribution of values of `type`.



High level model performance

Input data points can be divided into different categories based on their observed and predicted label. For instance, a **False Negative (FN)** is an input with a positive observed label (price > 1500) but negative predicted label (price <= 1500). A **True Negative (TN)** is an input whose observed and predicted labels are both negative. **True Positives (TP)** and **False Positives (FP)** are defined similarly.

Based on the model predictions, the inputs can be divided into different categories as:

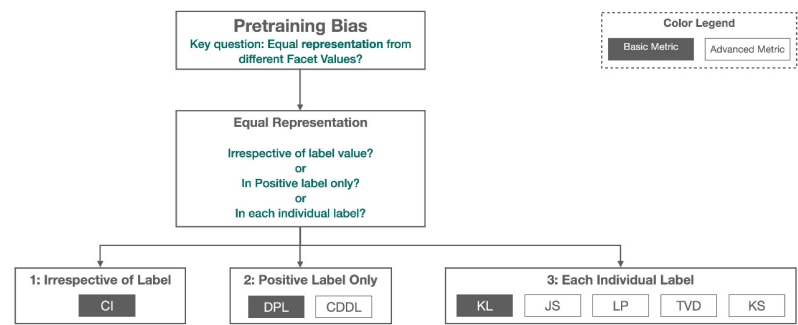


Here are metrics showing the model performance.

Metric	Description	Value
Accuracy	Proportion of inputs assigned the correct predicted label by the model.	0.944
Proportion of Positive Predictions in Labels	Proportion of input assigned in positive predicted label.	0.578
Proportion of Negative Predictions in Labels	Proportion of input assigned the negative predicted label.	0.422
True Positive Rate / Recall	Proportion of inputs with positive observed label correctly assigned the positive predicted label.	0.960
True Negative Rate / Specificity	Proportion of inputs with negative observed label correctly assigned the negative predicted label.	0.922
Acceptance Rate / Precision	Proportion of inputs with positive predicted label that actually have a positive observed label.	0.942
Rejection Rate	Proportion of inputs with negative predicted label that actually have a negative observed label.	0.947
Conditional Acceptance	Ratio between the positive observed labels and positive predicted labels.	0.981
Conditional Rejection	Ratio between the negative observed labels and negative predicted labels.	1.027
F1 Score	Harmonic mean of precision and recall.	0.951

Pre-training Bias Metrics

Pretraining bias metrics measure imbalances in facet value representation in the training data. Imbalances can be measured across different dimensions. For instance, you could focus imbalances within the inputs with positive observed label only. The figure below shows how different pretraining bias metrics focus on different dimensions. For a detailed description of these dimensions, see [Learn How Amazon SageMaker Clarify Helps Detect Bias](#).

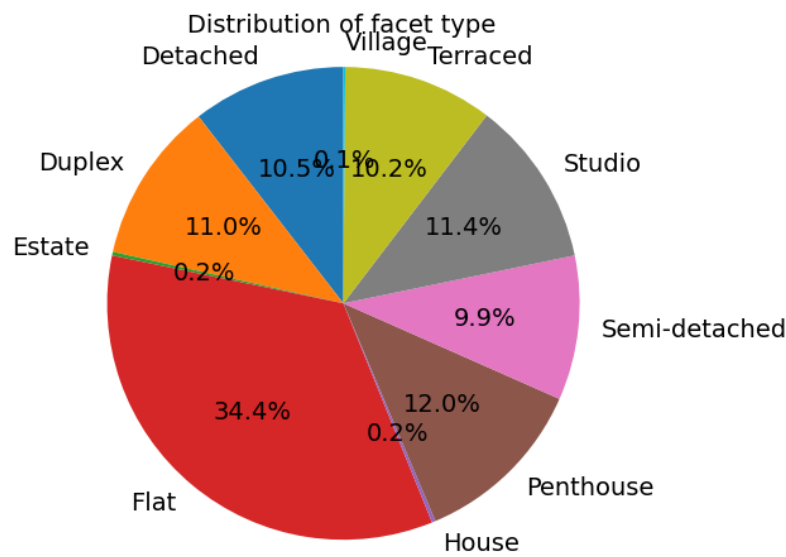


The metric values along with an informal description of what they mean are shown below. For mathematical formulas and examples, see the [Measure Pretraining Bias](https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-measure-data-bias.html) section of the AWS documentation.

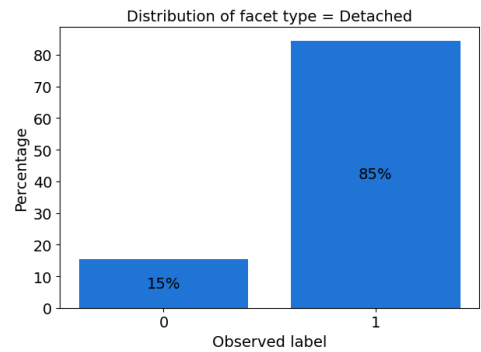
We computed the bias metrics for the label `price` using label value(s)/threshold `price > 1500` for the following facets:

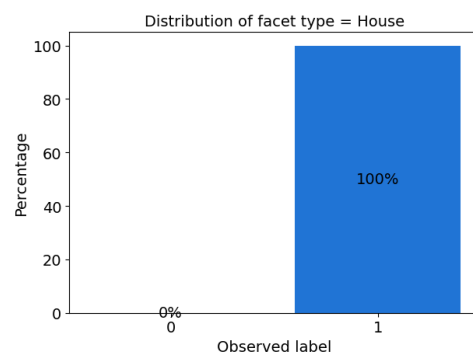
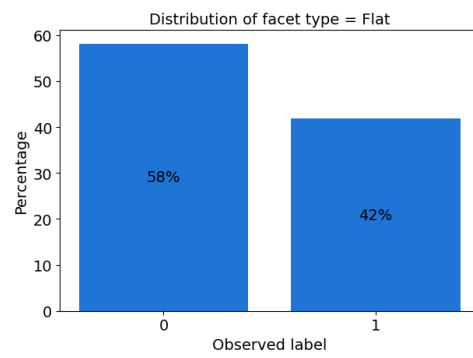
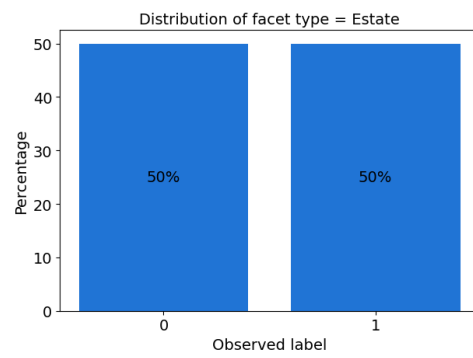
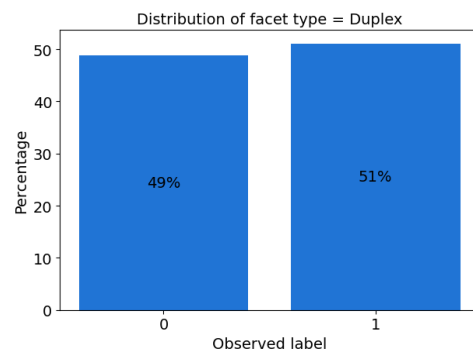
- Facet column: `type`

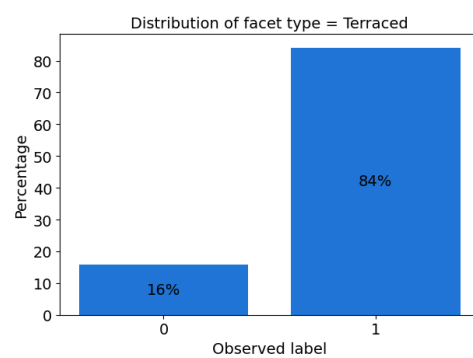
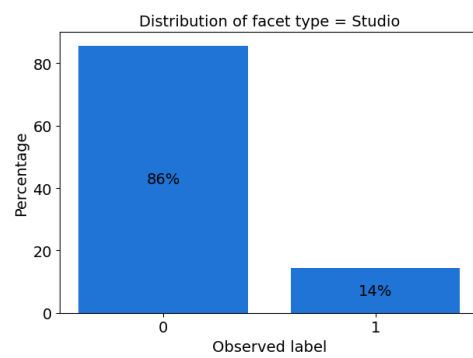
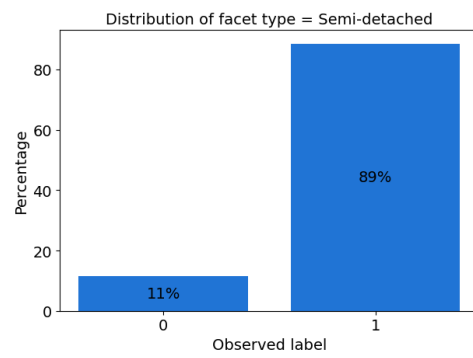
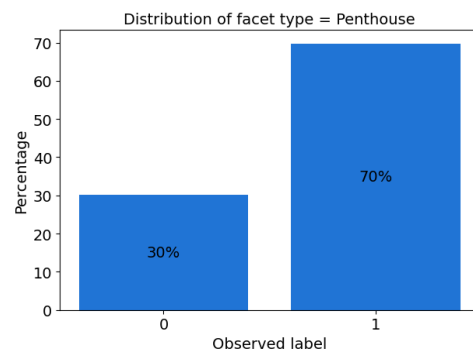
The pie chart shows the distribution of facet column `type` in your data.

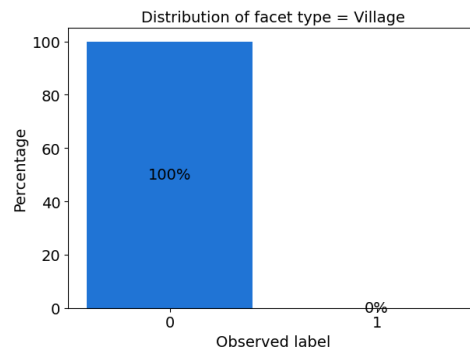


The bar plot(s) below show the distribution of facet column `type` in your data.









Facet Value(s)/Threshold: type = Detached

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values type = Detached and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values type = Detached and rest of the inputs.	0.790	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values type = Detached and rest of the inputs.	-0.312	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values type = Detached and rest of the inputs diverge from each other entropically.	0.059	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values type = Detached and rest of the inputs diverge from each other entropically.	0.269	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values type = Detached and rest of the inputs in the dataset.	0.312	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values type = Detached rest of the inputs in the dataset.	0.441	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values type = Detached and rest of the inputs in the dataset.	0.312	None

Facet Value(s)/Threshold: type = Duplex

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values type = Duplex and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values type = Duplex and rest of the inputs.	0.780	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values type = Duplex and rest of the inputs.	0.062	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values type = Duplex and rest of the inputs diverge from each other entropically.	0.002	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values type = Duplex and rest of the inputs diverge from each other entropically.	0.008	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values type = Duplex and rest of the inputs in the dataset.	0.062	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values type = Duplex rest of the inputs in the dataset.	0.087	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values type = Duplex and rest of the inputs in the dataset.	0.062	None

Facet Value(s)/Threshold: type = Estate

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values <code>type = Estate</code> and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values <code>type = Estate</code> and rest of the inputs.	0.995	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values <code>type = Estate</code> and rest of the inputs.	0.066	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values <code>type = Estate</code> and rest of the inputs diverge from each other entropically.	0.002	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values <code>type = Estate</code> and rest of the inputs diverge from each other entropically.	0.009	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values <code>type = Estate</code> and rest of the inputs in the dataset.	0.066	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values <code>type = Estate</code> rest of the inputs in the dataset.	0.094	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values <code>type = Estate</code> and rest of the inputs in the dataset.	0.066	None

Facet Value(s)/Threshold: `type = Flat`

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values <code>type = Flat</code> and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values <code>type = Flat</code> and rest of the inputs.	0.312	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values <code>type = Flat</code> and rest of the inputs.	0.226	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values <code>type = Flat</code> and rest of the inputs diverge from each other entropically.	0.026	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values <code>type = Flat</code> and rest of the inputs diverge from each other entropically.	0.103	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values <code>type = Flat</code> and rest of the inputs in the dataset.	0.226	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values <code>type = Flat</code> rest of the inputs in the dataset.	0.319	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values <code>type = Flat</code> and rest of the inputs in the dataset.	0.226	None

Facet Value(s)/Threshold: `type = House`

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values <code>type = House</code> and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values <code>type = House</code> and rest of the inputs.	0.995	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values <code>type = House</code> and rest of the inputs.	-0.435	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values <code>type = House</code> and rest of the inputs diverge from each other entropically.	0.031	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values <code>type = House</code> and rest of the inputs diverge from each other entropically.	-0.323	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values <code>type = House</code> and rest of the inputs in the dataset.	0.435	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values <code>type = House</code> rest of the inputs in the dataset.	0.435	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values <code>type = House</code> and rest of the inputs in the dataset.	0.217	None

Facet Value(s)/Threshold: `type = Penthouse`

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values type = Penthouse and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values type = Penthouse and rest of the inputs.	0.760	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values type = Penthouse and rest of the inputs.	-0.150	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values type = Penthouse and rest of the inputs diverge from each other entropically.	0.012	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values type = Penthouse and rest of the inputs diverge from each other entropically.	0.049	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values type = Penthouse and rest of the inputs in the dataset.	0.150	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values type = Penthouse rest of the inputs in the dataset.	0.212	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values type = Penthouse and rest of the inputs in the dataset.	0.150	None

Facet Value(s)/Threshold: type = Semi-detached

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values type = Semi-detached and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values type = Semi-detached and rest of the inputs.	0.802	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values type = Semi-detached and rest of the inputs.	-0.355	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values type = Semi-detached and rest of the inputs diverge from each other entropically.	0.080	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values type = Semi-detached and rest of the inputs diverge from each other entropically.	0.391	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values type = Semi-detached and rest of the inputs in the dataset.	0.355	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values type = Semi-detached rest of the inputs in the dataset.	0.502	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values type = Semi-detached and rest of the inputs in the dataset.	0.355	None

Facet Value(s)/Threshold: type = Studio

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values <code>type = Studio</code> and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values <code>type = Studio</code> and rest of the inputs.	0.772	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values <code>type = Studio</code> and rest of the inputs.	0.478	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values <code>type = Studio</code> and rest of the inputs diverge from each other entropically.	0.128	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values <code>type = Studio</code> and rest of the inputs diverge from each other entropically.	0.602	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values <code>type = Studio</code> and rest of the inputs in the dataset.	0.478	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values <code>type = Studio</code> rest of the inputs in the dataset.	0.676	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values <code>type = Studio</code> and rest of the inputs in the dataset.	0.478	None

Facet Value(s)/Threshold: `type = Terraced`

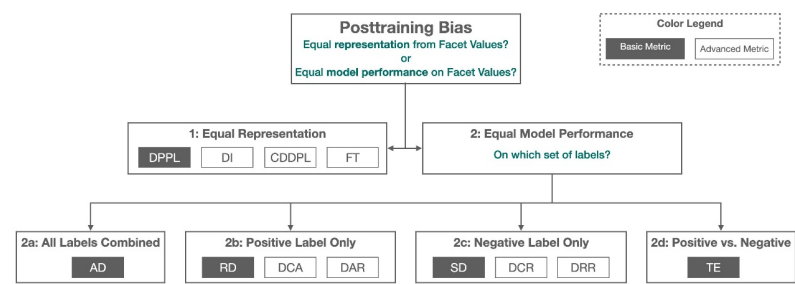
Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values <code>type = Terraced</code> and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values <code>type = Terraced</code> and rest of the inputs.	0.795	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values <code>type = Terraced</code> and rest of the inputs.	-0.307	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values <code>type = Terraced</code> and rest of the inputs diverge from each other entropically.	0.057	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values <code>type = Terraced</code> and rest of the inputs diverge from each other entropically.	0.258	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values <code>type = Terraced</code> and rest of the inputs in the dataset.	0.307	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values <code>type = Terraced</code> rest of the inputs in the dataset.	0.434	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values <code>type = Terraced</code> and rest of the inputs in the dataset.	0.307	None

Facet Value(s)/Threshold: `type = Village`

Metric	Description	Value	Error
Conditional Demographic Disparity in Labels (CDDL)	Measures maximum divergence between the observed label distributions for facet values <code>type = Village</code> and rest of the inputs in the dataset.	None	Error: see Clarify job output
Class Imbalance (CI)	Measures the imbalance in the number of inputs with facet values <code>type = Village</code> and rest of the inputs.	0.998	None
Difference in Proportions of Labels (DPL)	Measures the imbalance of positive observed labels between facet values <code>type = Village</code> and rest of the inputs.	0.567	None
Jensen-Shannon Divergence (JS)	Measures how much the observed label distributions of facet values <code>type = Village</code> and rest of the inputs diverge from each other entropically.	0.058	None
Kullback-Leibler Divergence (KL)	Measures how much the observed label distributions of facet values <code>type = Village</code> and rest of the inputs diverge from each other entropically.	-0.362	None
Kolmogorov-Smirnov (KS)	Measures maximum divergence between the observed label distributions for facet values <code>type = Village</code> and rest of the inputs in the dataset.	0.567	None
Lp-norm (LP)	Measures a p-norm difference between the observed label distributions associated with facet values <code>type = Village</code> rest of the inputs in the dataset.	0.567	None
Total Variation Distance (TVD)	Measures half of the L1-norm difference between the observed label distributions associated with facet values <code>type = Village</code> and rest of the inputs in the dataset.	0.283	None

Post-training Bias Metrics

Posttraining bias metrics measure imbalances in model predictions across different inputs. The figure below shows how different posttraining metrics target different types of imbalances over inputs. For a detailed description of these types, see [Learn How Amazon SageMaker Clarify Helps Detect Bias](#).

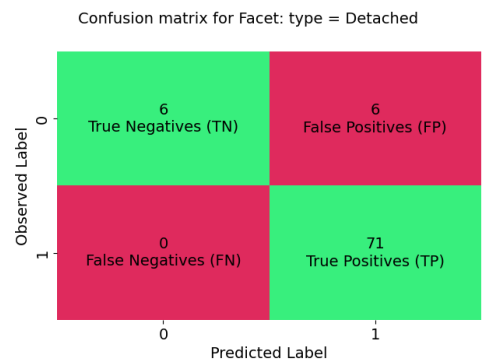


Bias can also result form imbalances in the model outcomes even when the facet value is not considered. The metric computing these imbalances is GE. The metric values along with an informal description of what they mean are shown below. For mathematical formulas and examples, see the [Measure Posttraining Data and Model Bias] (<https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-measure-post-training-bias.html>) section of the AWS documentation.

We computed the bias metrics for the label `price` using label value(s)/threshold `price > 1500` for the following facets:

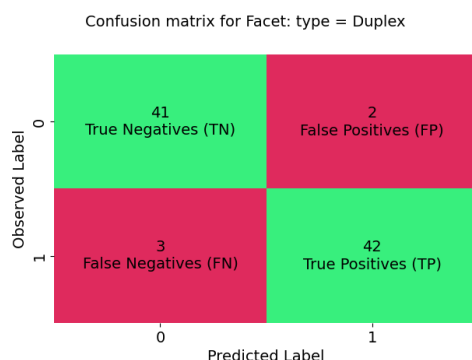
- Facet column: `type`

Facet Value(s)/Threshold: `type = Detached`



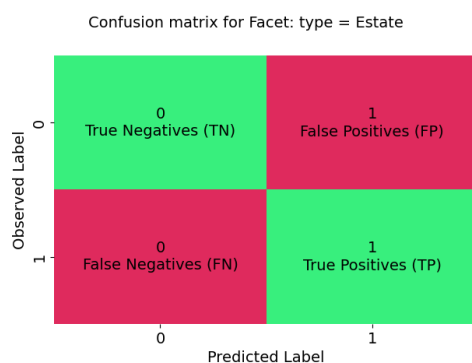
Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = Detached and rest of the inputs.	0.030	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = Detached and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = Detached and rest of the inputs.	0.038	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = Detached and rest of the inputs for predicted positive outcomes (acceptances).	0.085	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = Detached and rest of the inputs for negative outcomes (rejections).	1.161	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = Detached and rest of the inputs.	1.731	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = Detached and rest of the inputs.	-0.392	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = Detached and rest of the inputs.	0.054	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = Detached and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = Detached and rest of the inputs.	-0.047	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = Detached and rest of the inputs.	-0.479	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = Detached and rest of the inputs.	-0.900	None

Facet Value(s)/Threshold: type = Duplex



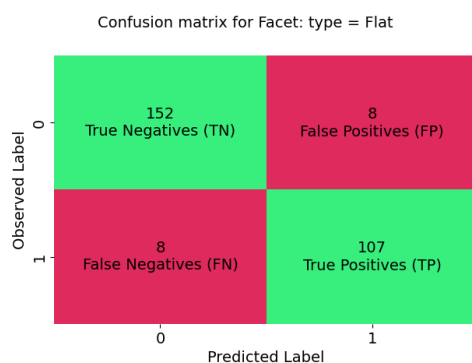
Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = Duplex and rest of the inputs.	0.001	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = Duplex and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = Duplex and rest of the inputs.	-0.014	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = Duplex and rest of the inputs for predicted positive outcomes (acceptances).	-0.047	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = Duplex and rest of the inputs for negative outcomes (rejections).	-0.057	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = Duplex and rest of the inputs.	0.852	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = Duplex and rest of the inputs.	0.087	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = Duplex and rest of the inputs.	-0.017	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = Duplex and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = Duplex and rest of the inputs.	0.030	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = Duplex and rest of the inputs.	0.036	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = Duplex and rest of the inputs.	0.900	None

Facet Value(s)/Threshold: type = Estate



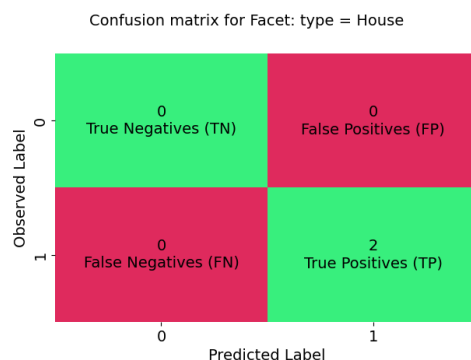
Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = Estate and rest of the inputs.	0.445	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = Estate and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = Estate and rest of the inputs.	0.443	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = Estate and rest of the inputs for predicted positive outcomes (acceptances).	0.483	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = Estate and rest of the inputs for negative outcomes (rejections).	Infinity	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = Estate and rest of the inputs.	1.735	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = Estate and rest of the inputs.	-0.424	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = Estate and rest of the inputs.	-0.947	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = Estate and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = Estate and rest of the inputs.	-0.040	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = Estate and rest of the inputs.	-0.925	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = Estate and rest of the inputs.	-0.692	None

Facet Value(s)/Threshold: type = Flat



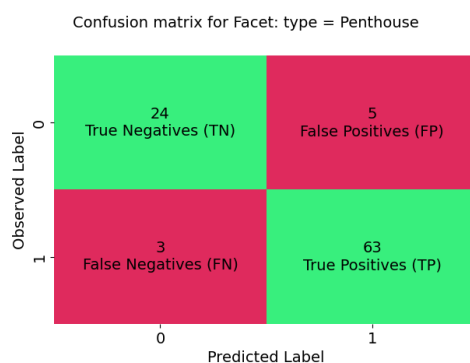
Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = Flat and rest of the inputs.	0.003	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = Flat and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = Flat and rest of the inputs.	0.015	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = Flat and rest of the inputs for predicted positive outcomes (acceptances).	-0.026	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = Flat and rest of the inputs for negative outcomes (rejections).	-0.051	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = Flat and rest of the inputs.	0.633	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = Flat and rest of the inputs.	0.243	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = Flat and rest of the inputs.	0.006	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = Flat and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = Flat and rest of the inputs.	0.040	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = Flat and rest of the inputs.	0.052	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = Flat and rest of the inputs.	0.474	None

Facet Value(s)/Threshold: type = House



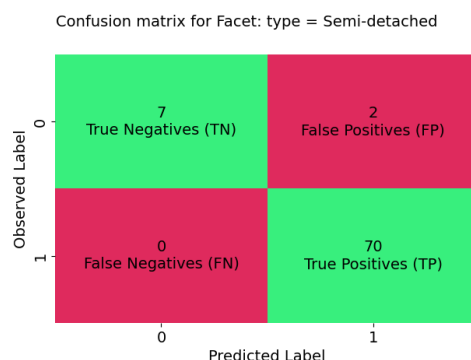
Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = House and rest of the inputs.	-0.056	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = House and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = House and rest of the inputs.	-0.059	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = House and rest of the inputs for predicted positive outcomes (acceptances).	-0.020	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = House and rest of the inputs for negative outcomes (rejections).	-1.027	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = House and rest of the inputs.	1.735	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = House and rest of the inputs.	-0.424	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = House and rest of the inputs.	-0.947	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = House and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = House and rest of the inputs.	-0.040	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = House and rest of the inputs.	-0.922	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = House and rest of the inputs.	-0.667	None

Facet Value(s)/Threshold: type = Penthouse



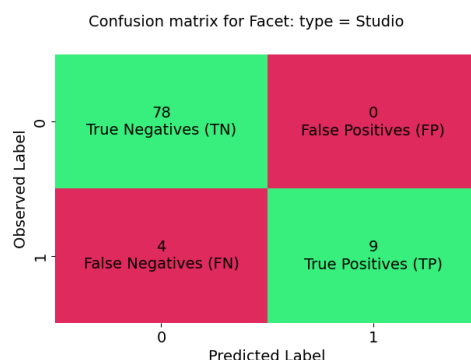
Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values <code>type = Penthouse</code> and rest of the inputs.	0.031	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values <code>type = Penthouse</code> and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values <code>type = Penthouse</code> and rest of the inputs.	0.016	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values <code>type = Penthouse</code> and rest of the inputs for predicted positive outcomes (acceptances).	0.011	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values <code>type = Penthouse</code> and rest of the inputs for negative outcomes (rejections).	0.052	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values <code>type = Penthouse</code> and rest of the inputs.	1.288	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values <code>type = Penthouse</code> and rest of the inputs.	-0.161	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values <code>type = Penthouse</code> and rest of the inputs.	-0.063	None
Counterfactual Fliptest (FT)	Examines each input with facet value <code>type = Penthouse</code> and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values <code>type = Penthouse</code> and rest of the inputs.	0.006	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values <code>type = Penthouse</code> and rest of the inputs.	-0.103	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values <code>type = Penthouse</code> and rest of the inputs.	-0.082	None

Facet Value(s)/Threshold: `type = Semi-detached`



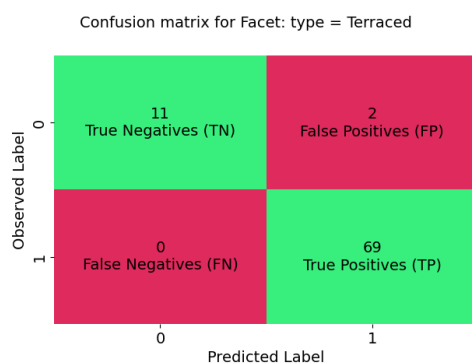
Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = Semi-detached and rest of the inputs.	-0.034	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = Semi-detached and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = Semi-detached and rest of the inputs.	-0.036	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = Semi-detached and rest of the inputs for predicted positive outcomes (acceptances).	0.010	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = Semi-detached and rest of the inputs for negative outcomes (rejections).	0.265	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = Semi-detached and rest of the inputs.	1.685	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = Semi-detached and rest of the inputs.	-0.370	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = Semi-detached and rest of the inputs.	0.054	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = Semi-detached and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = Semi-detached and rest of the inputs.	-0.047	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = Semi-detached and rest of the inputs.	-0.148	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = Semi-detached and rest of the inputs.	-0.720	None

Facet Value(s)/Threshold: type = Studio



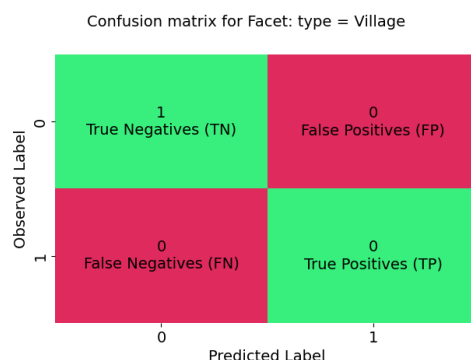
Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = Studio and rest of the inputs.	-0.014	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = Studio and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = Studio and rest of the inputs.	-0.060	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = Studio and rest of the inputs for predicted positive outcomes (acceptances).	-0.473	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = Studio and rest of the inputs for negative outcomes (rejections).	-0.100	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = Studio and rest of the inputs.	0.155	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = Studio and rest of the inputs.	0.540	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = Studio and rest of the inputs.	0.006	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = Studio and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = Studio and rest of the inputs.	0.276	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = Studio and rest of the inputs.	0.100	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = Studio and rest of the inputs.	Infinity	None

Facet Value(s)/Threshold: type = Terraced



Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = Terraced and rest of the inputs.	-0.035	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = Terraced and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = Terraced and rest of the inputs.	-0.036	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = Terraced and rest of the inputs for predicted positive outcomes (acceptances).	0.010	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = Terraced and rest of the inputs for negative outcomes (rejections).	0.160	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = Terraced and rest of the inputs.	1.590	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = Terraced and rest of the inputs.	-0.321	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = Terraced and rest of the inputs.	0.055	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = Terraced and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = Terraced and rest of the inputs.	-0.047	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = Terraced and rest of the inputs.	-0.079	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = Terraced and rest of the inputs.	-0.720	None

Facet Value(s)/Threshold: type = Village



Metric	Description	Value	Error
Accuracy Difference (AD)	Measures the difference between the prediction accuracy for facet values type = Village and rest of the inputs.	-0.056	None
Conditional Demographic Disparity in Predicted Labels (CDDPL)	Measures the disparity of predicted labels between facet values type = Village and rest of the inputs as a whole, but also by subgroups dictated by Age.	None	Error: see Clarify job output
Difference in Acceptance Rates (DAR)	Measures the difference in the ratios of the observed positive outcomes (TP) to the predicted positives (TP + FP) between facet values type = Village and rest of the inputs.	0.942	None
Difference in Conditional Acceptance (DCAcc)	Compares the observed labels to the labels predicted by the model. Assesses whether this is the same across facet values type = Village and rest of the inputs for predicted positive outcomes (acceptances).	0.981	None
Difference in Conditional Rejection (DCR)	Compares the observed labels to the labels predicted by the model and assesses whether this is the same across facet values type = Village and rest of the inputs for negative outcomes (rejections).	-0.027	None
Disparate Impact (DI)	Measures the ratio of proportions of the predicted labels for facet values type = Village and rest of the inputs.	0.000	None
Difference in Positive Proportions in Predicted Labels (DPPL)	Measures the difference in the proportion of positive predictions between facet values type = Village and rest of the inputs.	0.578	None
Difference in Rejection Rates (DRR)	Measures the difference in the ratios of the observed negative outcomes (TN) to the predicted negatives (TN + FN) between facet values type = Village and rest of the inputs.	0.053	None
Counterfactual Fliptest (FT)	Examines each input with facet value type = Village and assesses whether similar members from rest of the inputs have different model predictions.	None	Error: see Clarify job output
Generalized entropy (GE)	Measures the inequality in benefits b assigned to each input by the model predictions.	0.027	None
Recall Difference (RD)	Measures the difference between the recall, aka true positive rate, of the model for facet values type = Village and rest of the inputs.	0.960	None
Specificity difference (SD)	Measures the difference between the specificity, aka true negative rate, of the model for facet values type = Village and rest of the inputs.	0.078	None
Treatment Equality (TE)	Measures the difference in the ratio of false positives to false negatives between facet values type = Village and rest of the inputs.	-0.667	None

Appendix: Analysis Configuration Parameters

```
{
  "dataset_type": "text/csv",
  "label": "price",
  "label_values_or_threshold": [
    1500
  ],
  "facet": [
    {
      "name_or_index": "type"
    }
  ],
  "methods": {
    "report": {
      "name": "report",
      "title": "Analysis Report"
    },
    "pre_training_bias": {
      "methods": "all"
    },
    "post_training_bias": {
      "methods": "all"
    }
  },
  "predictor": {
    "model_name": "pre-trained model 20250615_101500"
```

```
    "model_name": "madrid-rent-model-20250615-101509",
    "instance_type": "ml.m5.large",
    "initial_instance_count": 1,
    "accept_type": "text/csv",
    "content_type": "text/csv"
  }
}
```